

**For CI MED Students, Class of 2029, Only-Do not Copy, Share,
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Multidisciplinary Review Session

Histology and Histopathology Annotations and Explanations

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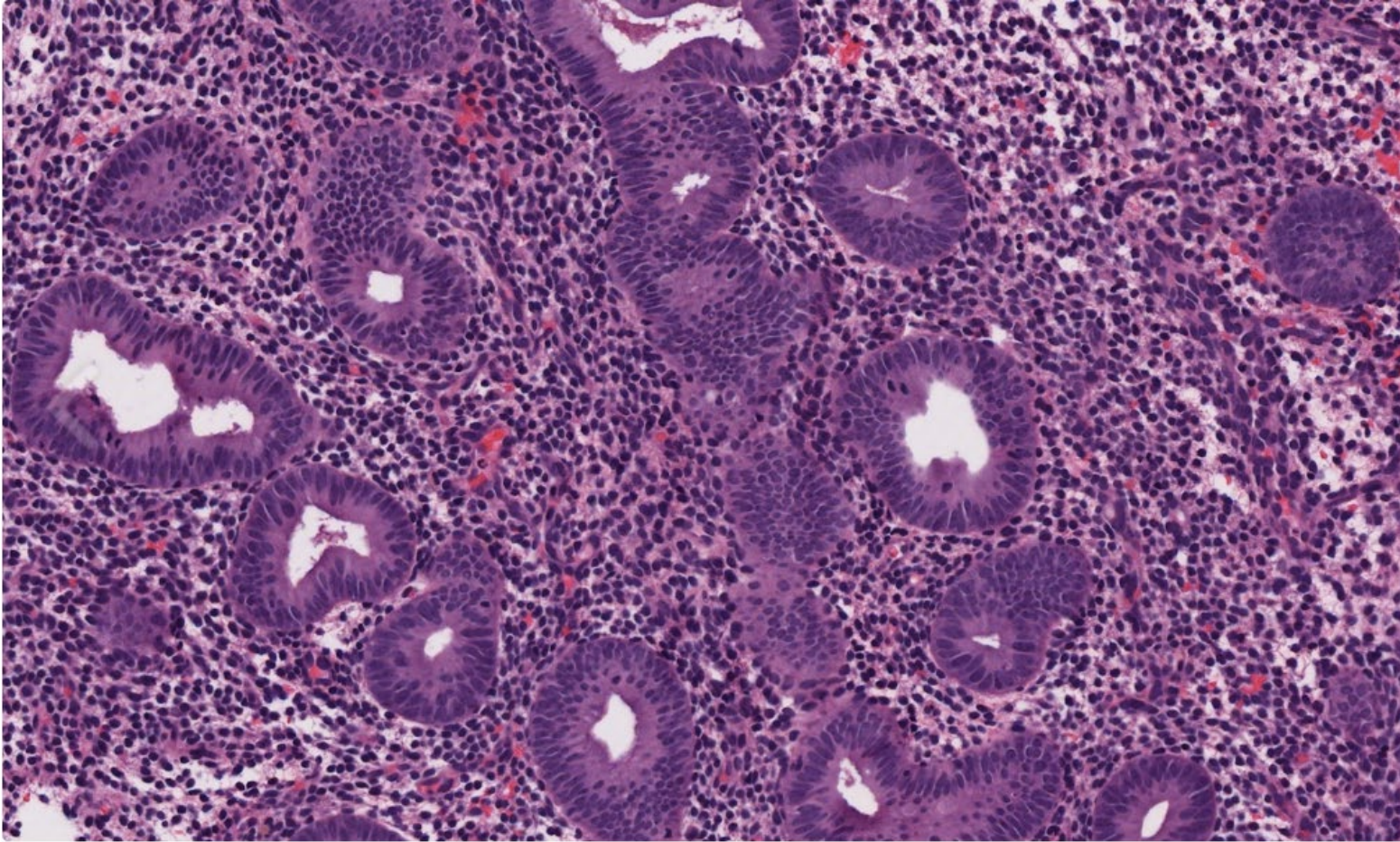
Pathology and Histology Discipline Lead

Carle Illinois College of Medicine

**Please refer to Histology and Pathology Lectures for more
explanations and practice questions**

Station 2 Histology and Histopathology

Station 2; Question 2



Proliferative Endometrium

Proliferative phase endometrium is characterized by round glands with pseudostratified cells and numerous mitotic figures in both glands and stroma.

Station 2: Question 3, Part 2

Match Histopathology with Uterine 3D-Printed Pathologies

1. Adenomyosis: Endometrial tissue (gland and stroma) invading the myometrium

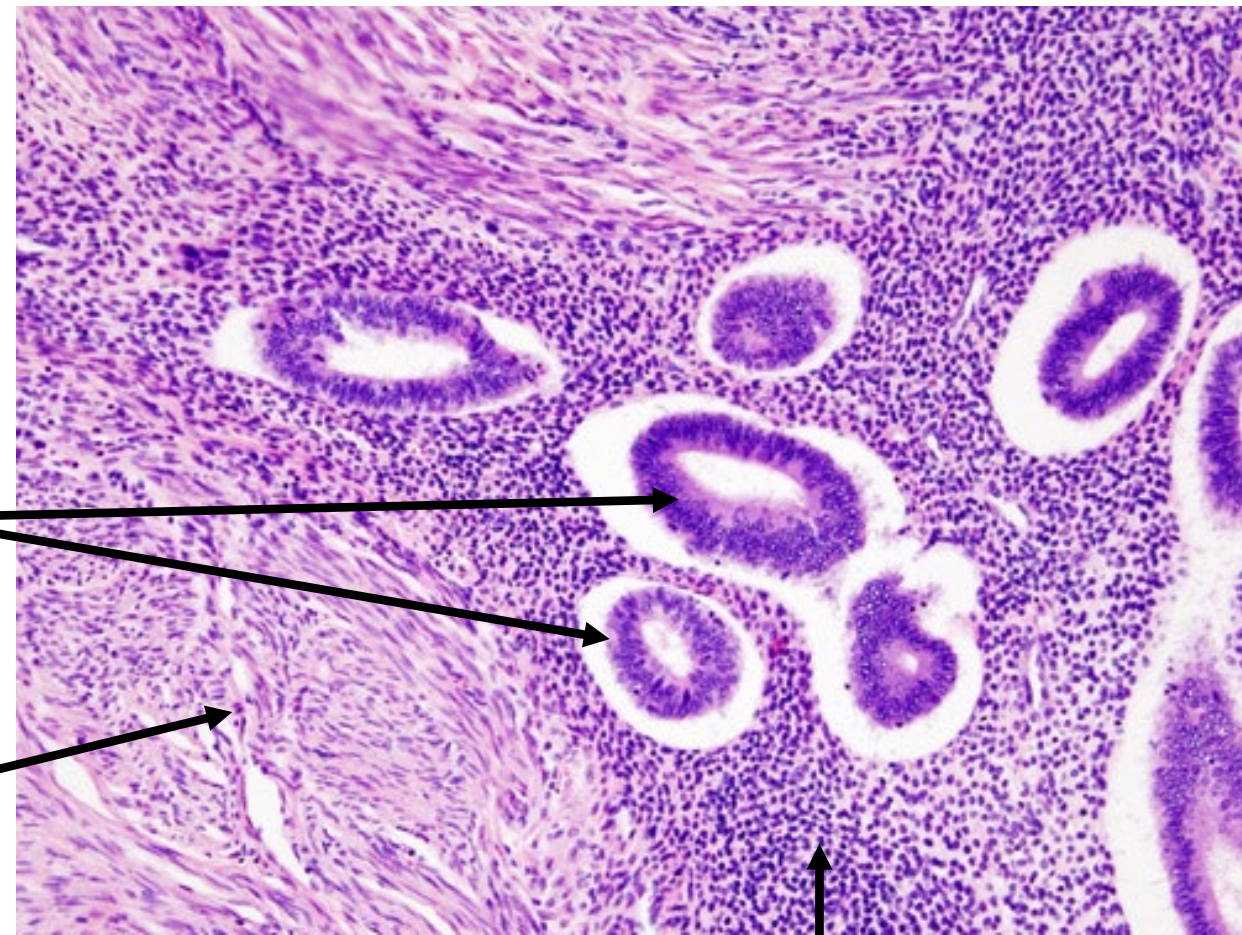
- This ectopic tissue induces reactive hypertrophy of surrounding smooth muscle
- Results in a globular, enlarged uterus with a thickened uterine wall
- The stromal component of adenomyosis is thought to be the major driver of reactive myometrial hypertrophy and hyperplasia through local inflammatory, estrogenic, and growth-factor signaling.



- Uterus appears enlarged and boggy
- Wall may be diffusely thickened
- Hemorrhagic cystic spaces representing ectopic endometrial tissue

Endometrial Glands

Myometrium



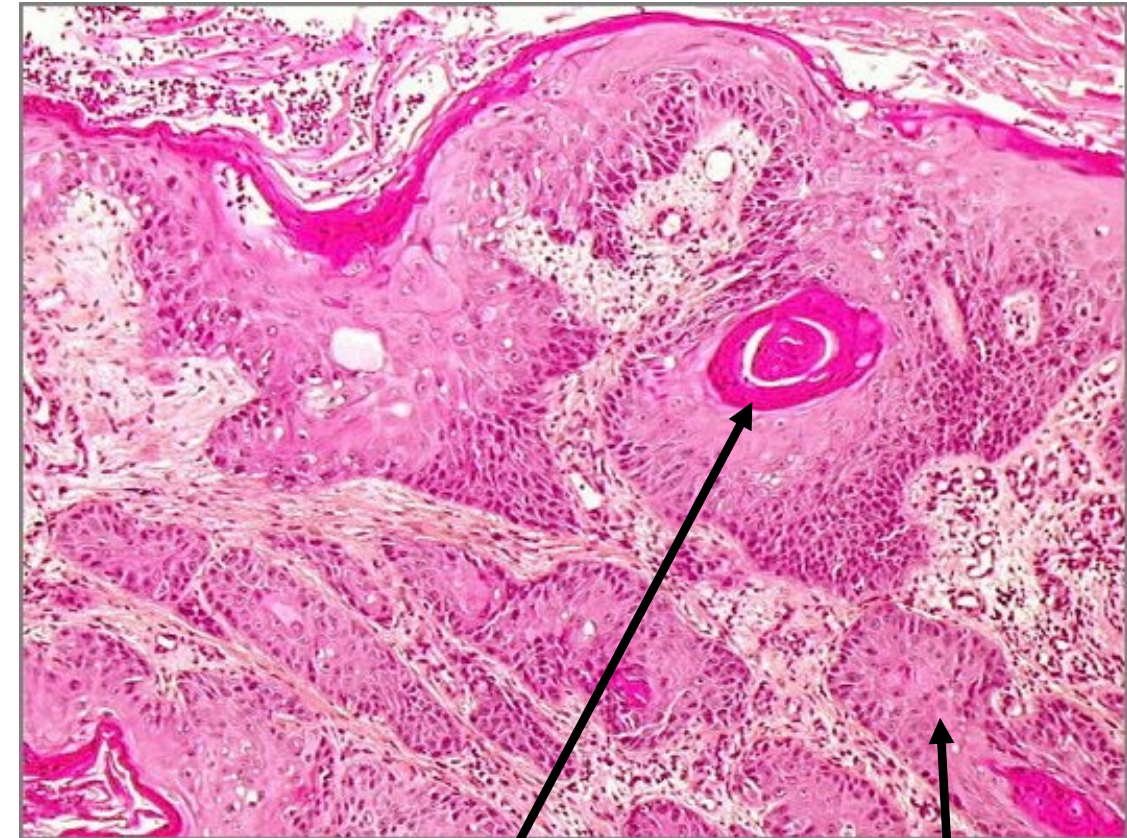
Endometrial Stromal Cells



2. Cervical SCC:

- Well-differentiated SCC of the cervix
- Infiltrating networks of bands of neoplastic cells invading stroma
- Malignant cells demonstrate keratinization and keratin pearls (whorls of epidermoid cells containing central nests of keratin)
- The nuclei are large and hyperchromatic with coarse chromatin.

Large squamous carcinoma (bottom of the picture) has obliterated the cervix and invaded the lower uterine segment with **areas of hemorrhage** and necrosis



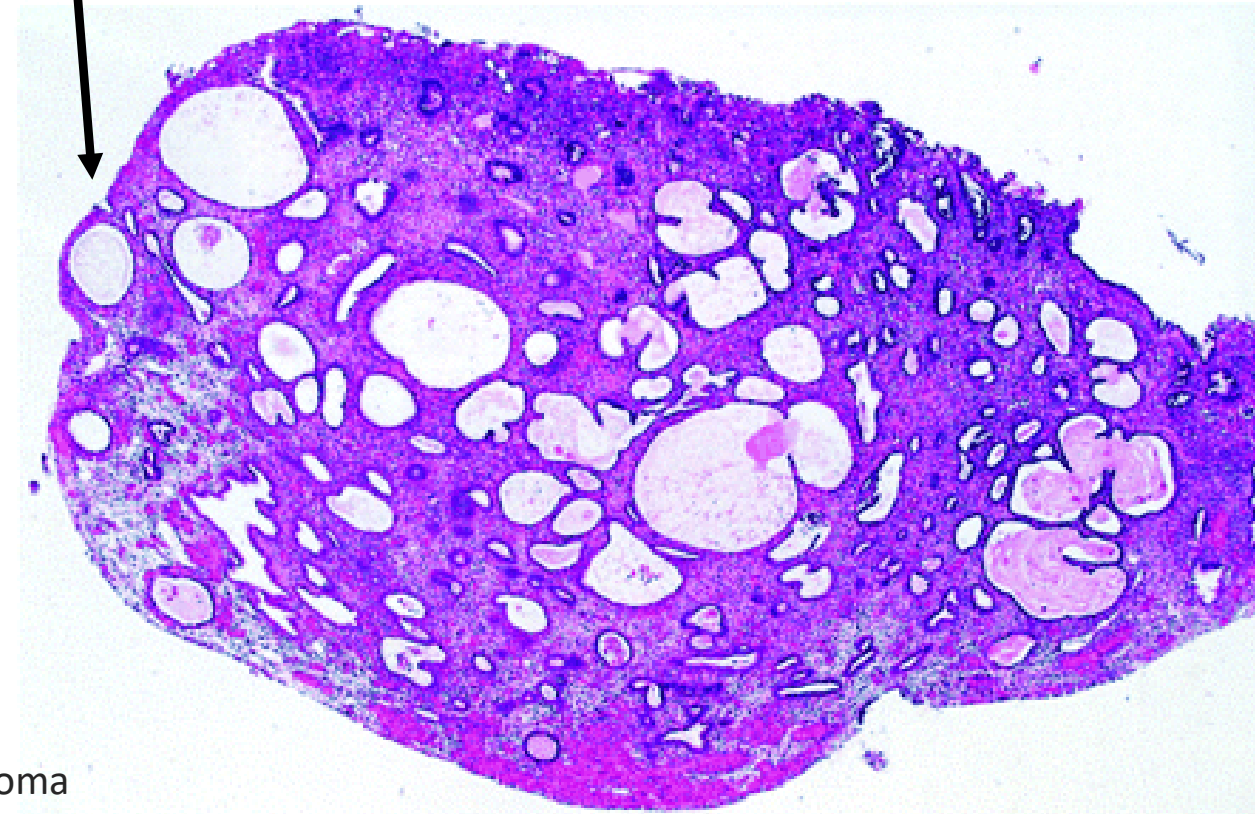
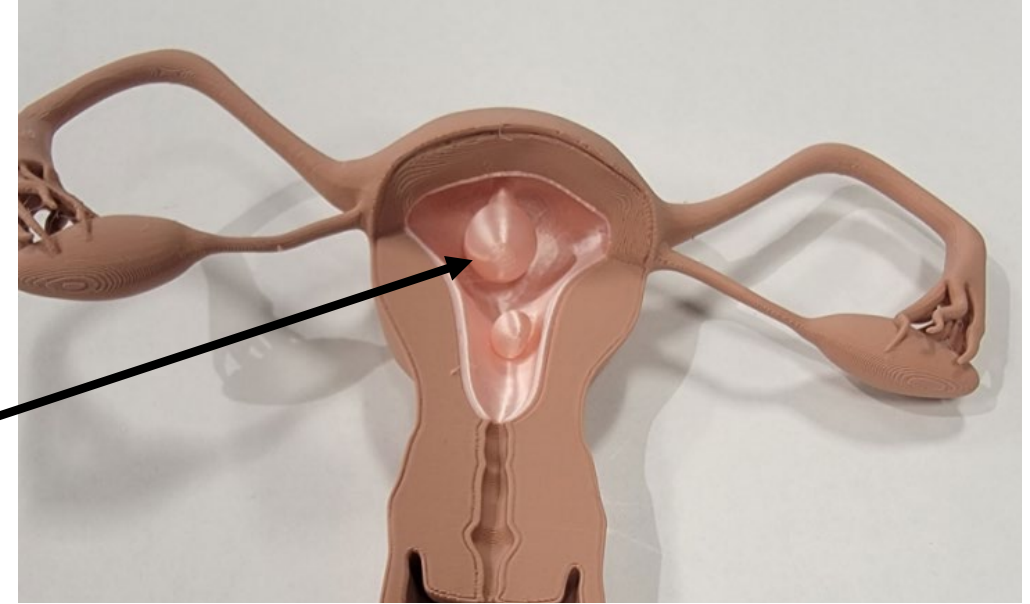
Keratin Pearls

Sheets of malignant squamous cells

3. Endometrial Polyp

- Endometrial **polyps** are abnormal growths of the endometrium
- either **sessile** (a non-mobile polyp with a broad base with no stalk) or **pedunculated** (projecting into the uterine cavity and attached by a stalk).
- Histologically, they resemble the **stratum basalis** of the endometrium (i.e., **cystically dilated** glands with small muscular arteries)
- The endometrial polyp **stroma** is more **collagenous** than the normal endometrial stroma (more cellular)
- diagnosed in the presence of 2 of the following 3:
 - Thick-walled vessels
 - Collagenous stroma
 - Epithelium on at least 3 sides
- Endometrial polyps usually present with abnormal uterine bleeding.
- Although malignant transformation is rare, Endometrial carcinoma can arise in endometrial polyps

Endometrial Polyp



simple cystic hyperplasia with dilated glands in loose stroma

4. Non-atypical Endometrial Hyperplasia:

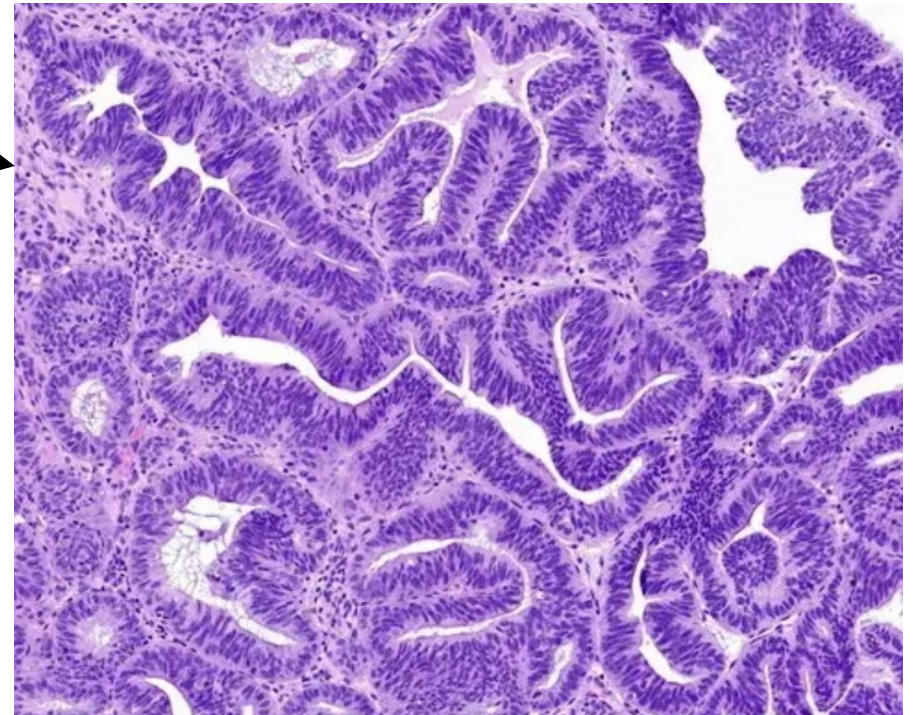
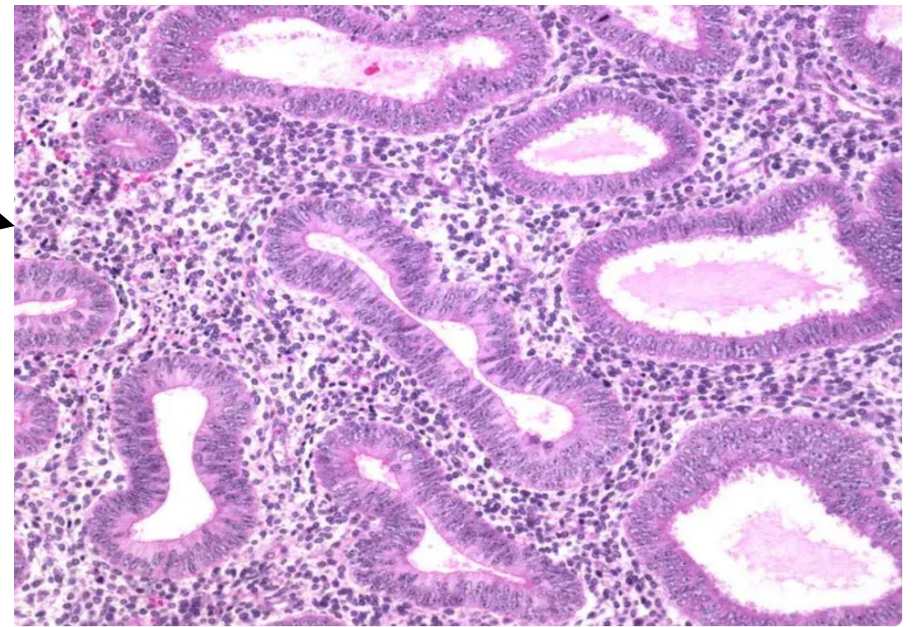
- Closely packed tubular, irregular or branching glands
- lined by pseudostratified columnar epithelium resembling proliferative phase endometrium.
- There is no cytologic atypia.

Compare To

Atypical Endometrial Hyperplasia:

Atypical Endometrial Hyperplasia/Endometrioid Intraepithelial Neoplasia (AEH/EIN): The **glands** are arranged **back-to-back** with little intervening stroma (gland-to-stroma ratio >1). The glandular profiles are **round and tubular** or have **complex, stellate or irregular shapes** with **branching** and **outpouchings**.

Cytologic atypia is characterized by **large round or elongated nuclei** with **high N:C ratio**, **loss of nuclear polarity** with irregular distribution of nuclei, and inconspicuous nucleoli. Some cases show distinct nucleoli. **Mitotic activity** is **variably increased**.



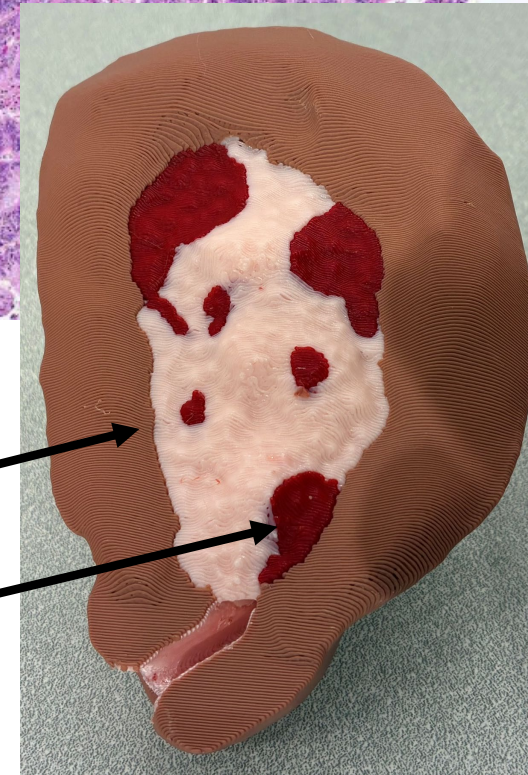
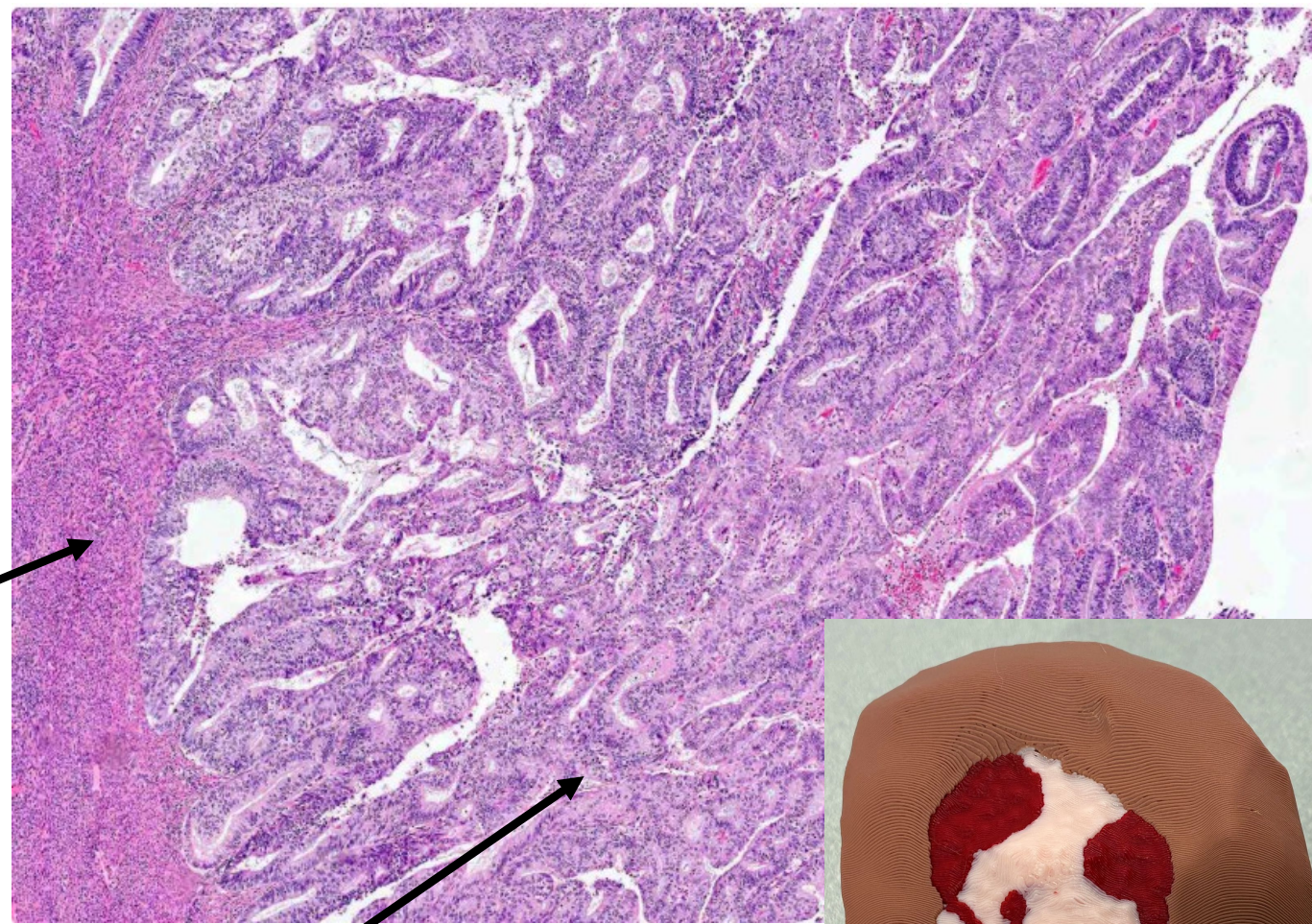
5. Endometrial Carcinoma

Grade 1 Endometrioid Endometrial Carcinoma: predominantly confluent (back-to-back) glandular to papilliform growth with no intervening stroma (DDX hyperplasia), cytologically resembling proliferative-type endometrium. Glands invade the myometrium

Myometrium invaded
by malignant glands

Back to back endometrial
glands without intervening
stroma

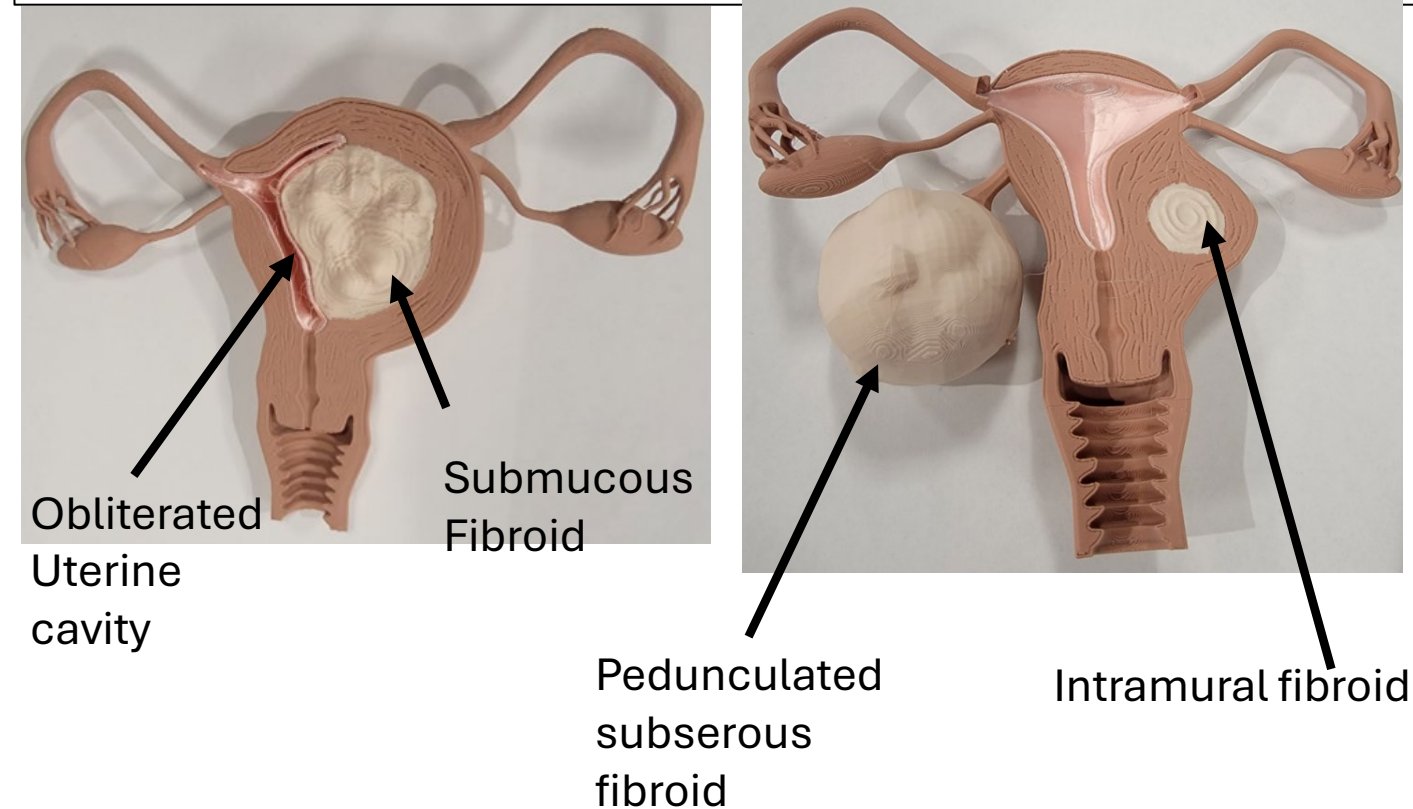
- **Carcinoma occupies most of the endometrial cavity** with **areas of hemorrhage**
- The cervix is unaffected



6. Leiomyoma (Fibroid)

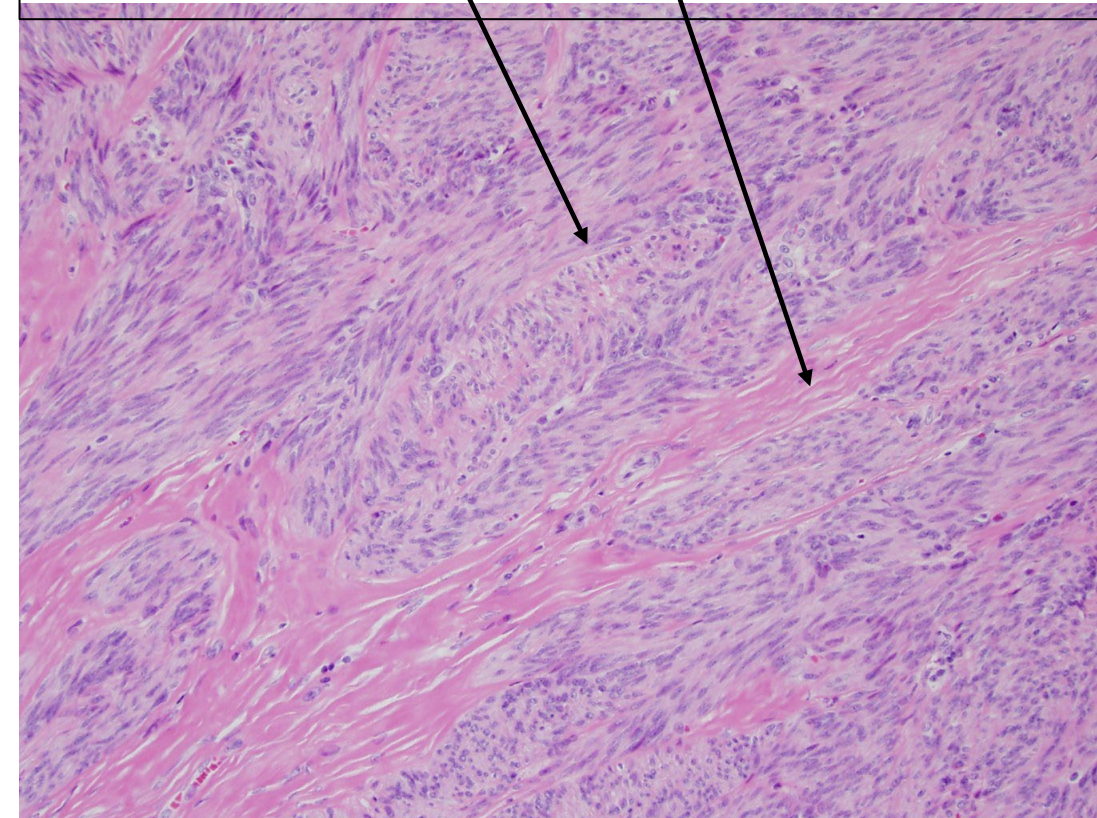
Gross:

- Sharply circumscribed, firm gray-white masses with a characteristic whorled cut surface
- Embedded within the myometrium (intramural), lie beneath the endometrium (submucosal) or the serosa (subserosal); may extend out on attenuated stalks (pedunculated)



Microscopic:

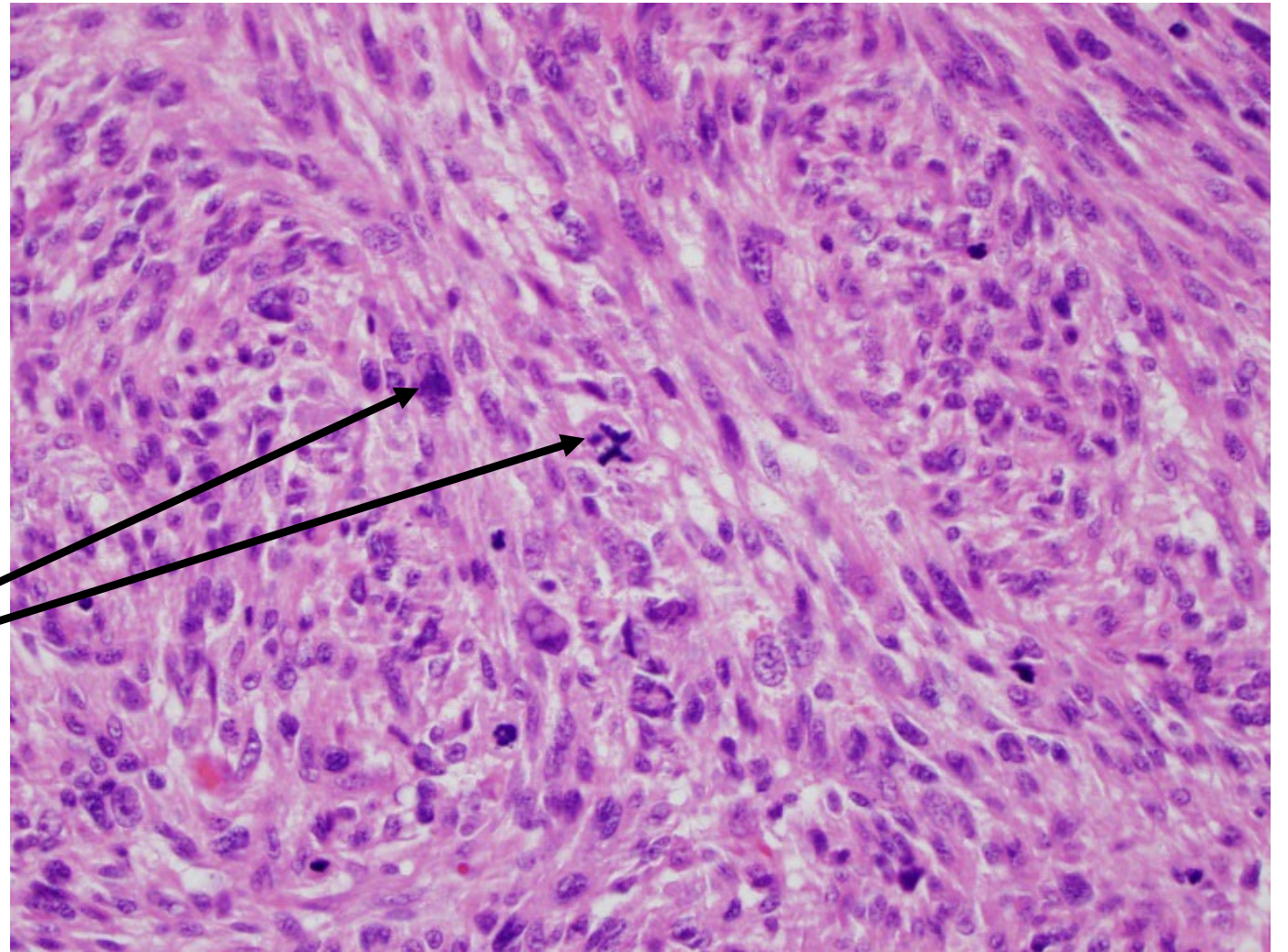
- Bundles of intersecting uniform **smooth muscle cells** (elongated nuclei and eosinophilic cytoplasm)
- Similar in appearance to typical myometrium
- Foci of fibrosis (collagen bundles), calcification, and degenerative softening may be present



7. Leiomyosarcoma

Leiomyosarcoma is a cellular tumor composed of fascicles of spindle cells exhibiting smooth muscle differentiation with moderate to severe pleomorphism. Usually coagulative tumor cell necrosis is present and abnormal mitoses exceed 10-15/10 HPF

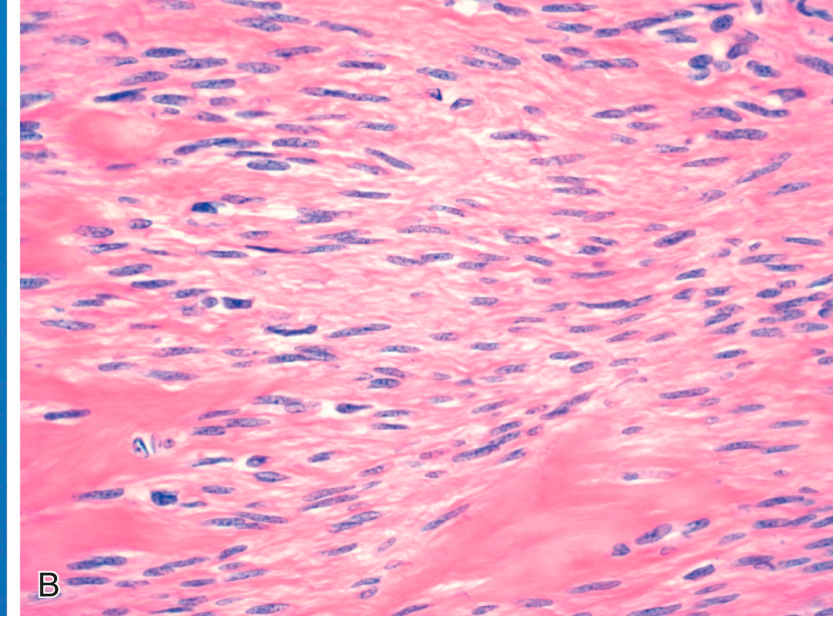
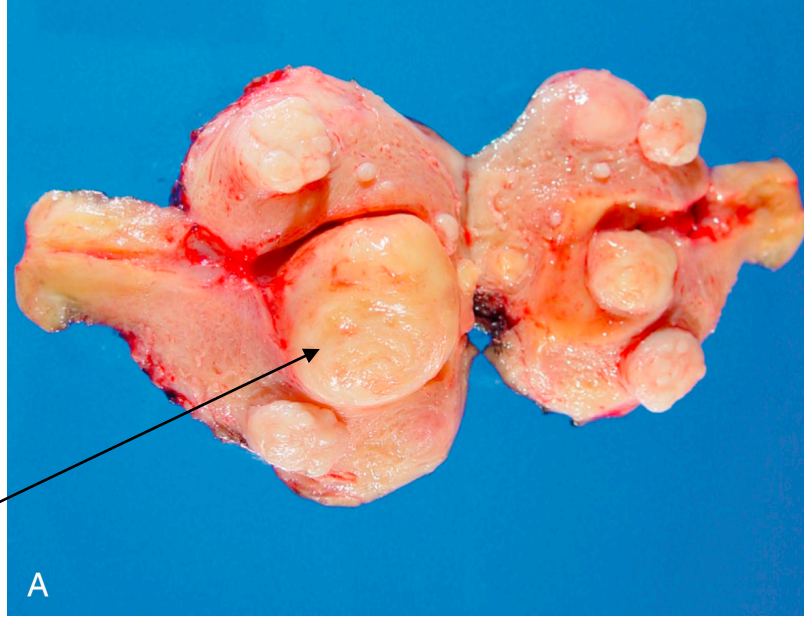
Abnormal Mitoses



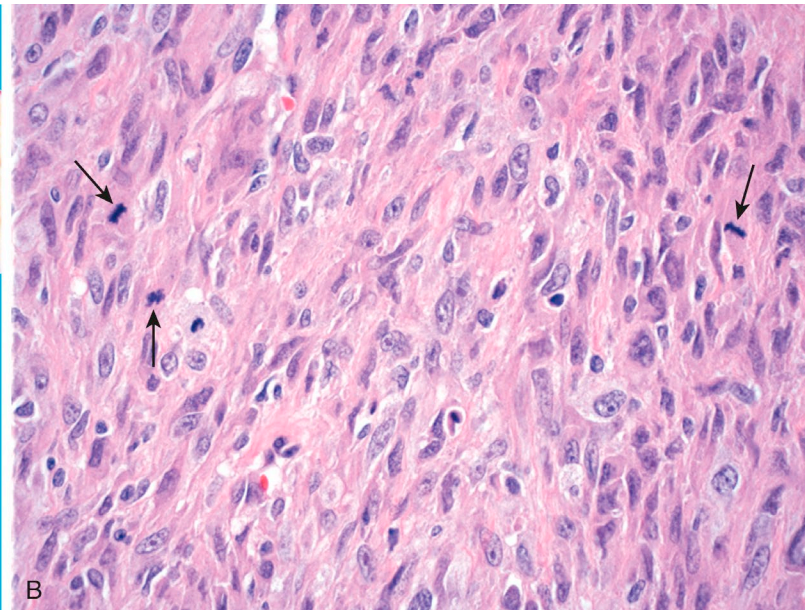
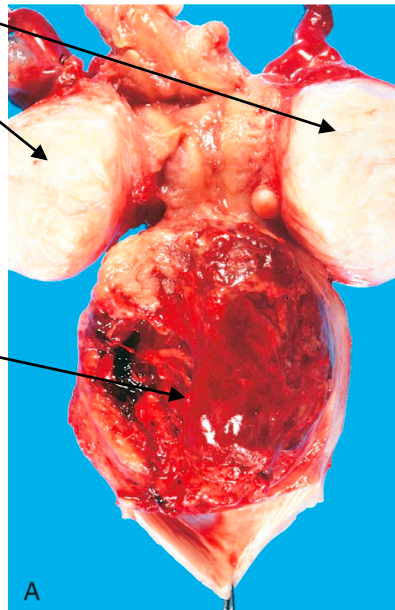
Leiomyoma vs Leiomyosarcoma

LO: Compare benign and malignant uterine tumors (e.g., leiomyomas vs. leiomyosarcomas), including morphology, clinical course, and prognosis.

Leiomyomas



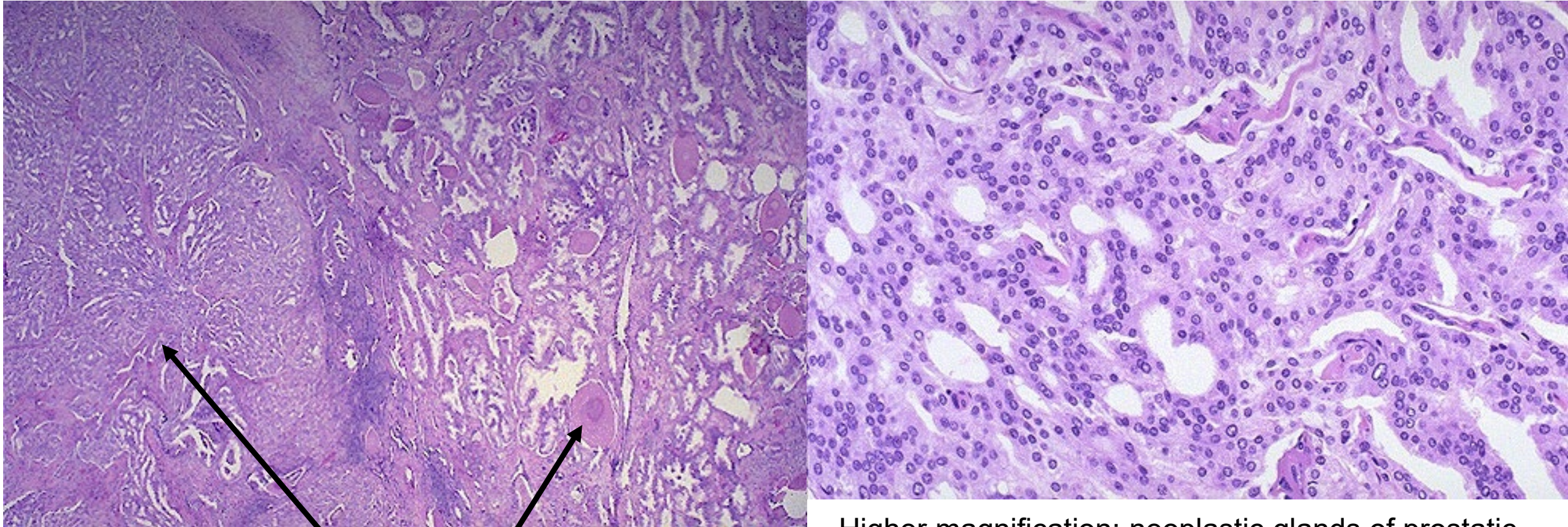
Leiomyosarcoma



Station 5: Question 4, Prostate Histology and Histopathology

Station 5, Question 4

Patient B- Prostate cancer

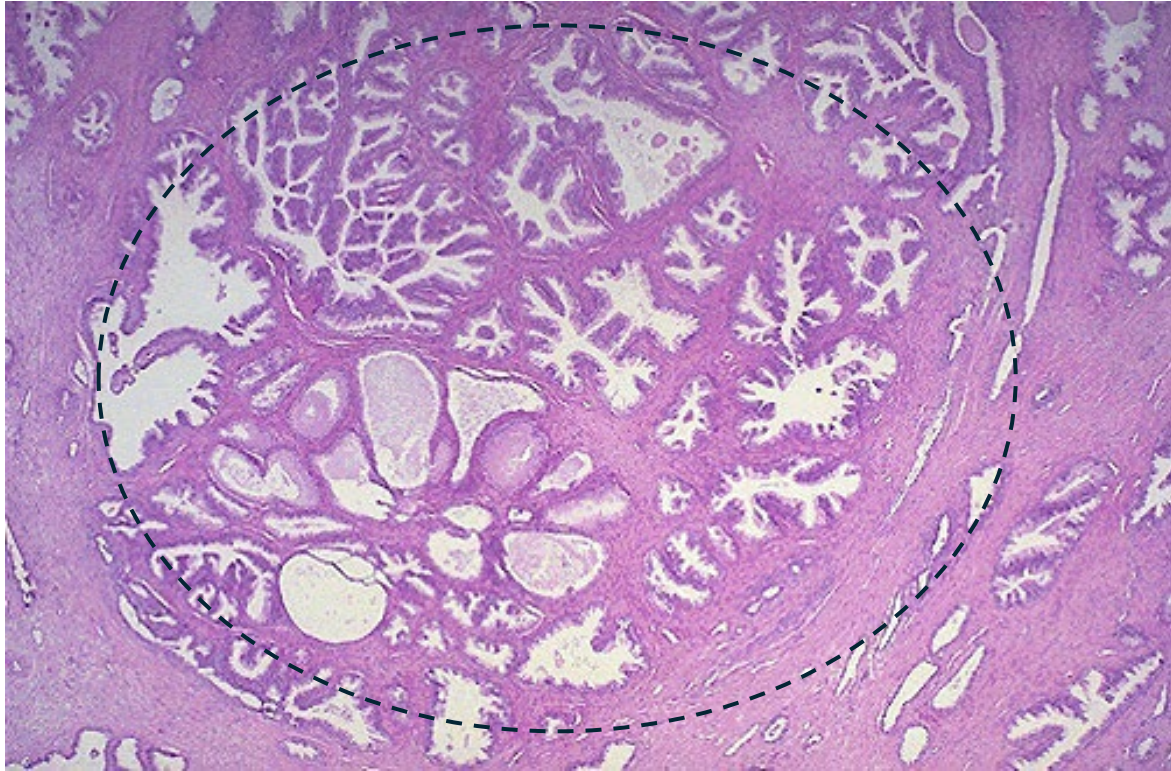


- On the left is **prostatic adenocarcinoma**; glands of the carcinoma are small and crowded.
- On the right: normal prostatic glands containing scattered **corpora amylacea**.

Higher magnification: neoplastic glands of prostatic adenocarcinoma are still recognizable as glands, **no intervening stroma and the nuclei are hyperchromatic.**

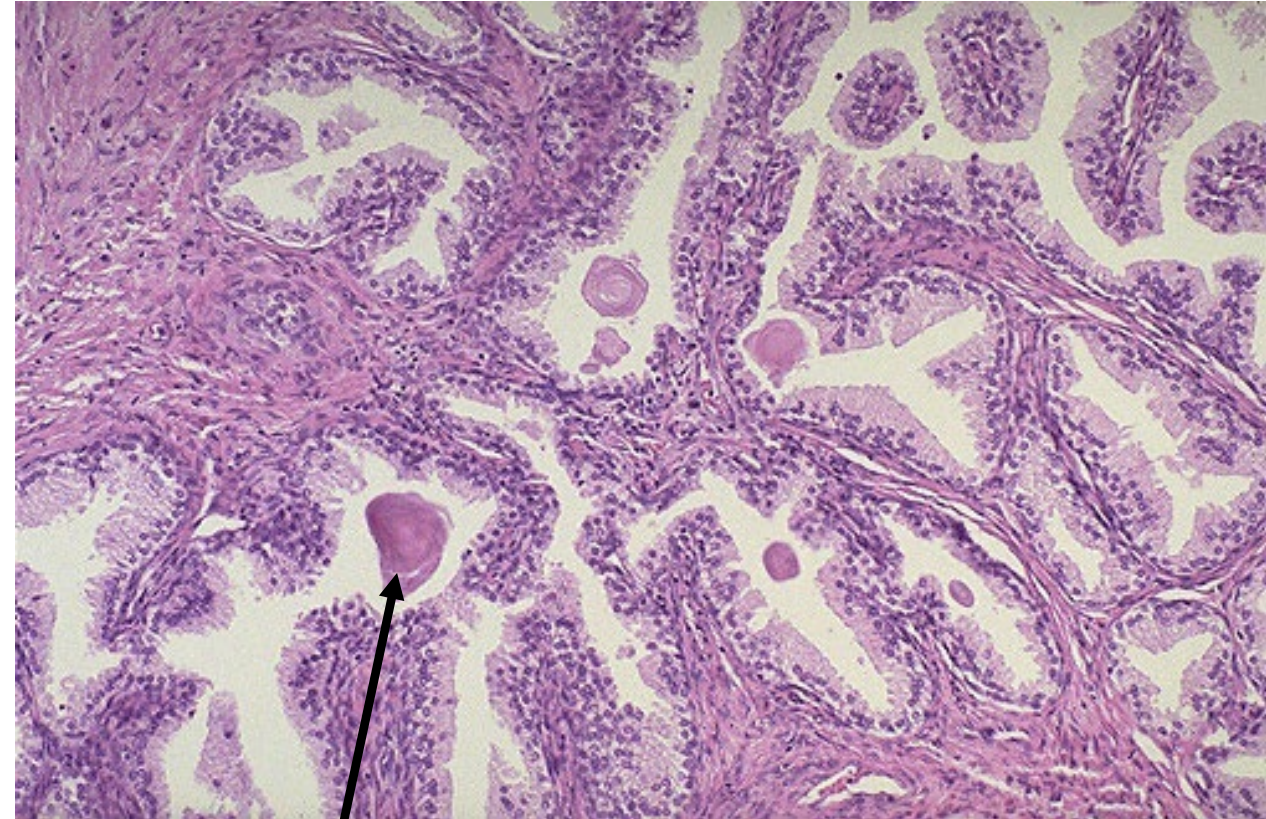
Station 5, Question 4

Patient A- BPH



Benign prostatic hyperplasia involves both glands and stroma

A large hyperplastic nodule of glands is seen



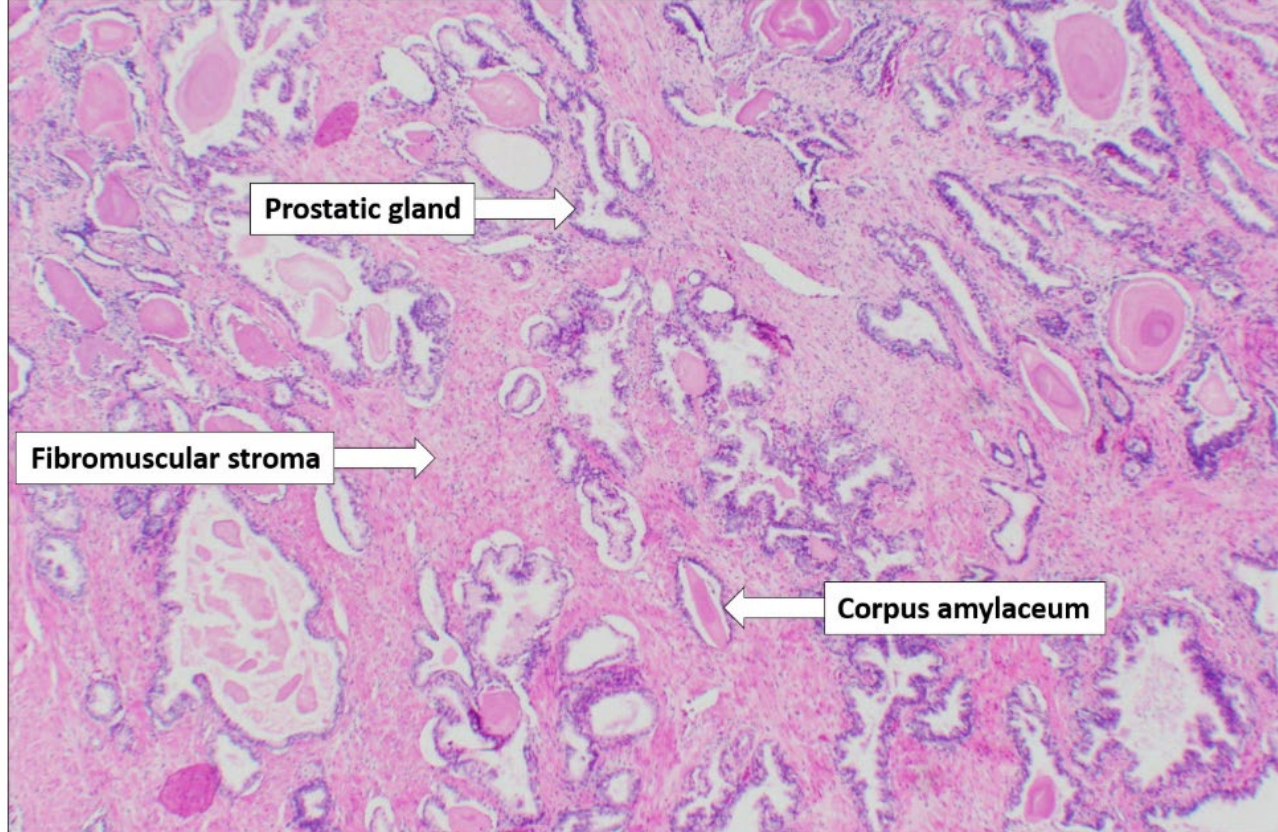
Higher magnification: the enlarged prostate has glandular hyperplasia. The glands are well-differentiated and **still have some intervening stroma**. The small laminated pink concretions within the glandular lumens are known as **corpora amylacea**.

Station 5, Question 4

Normal Prostate

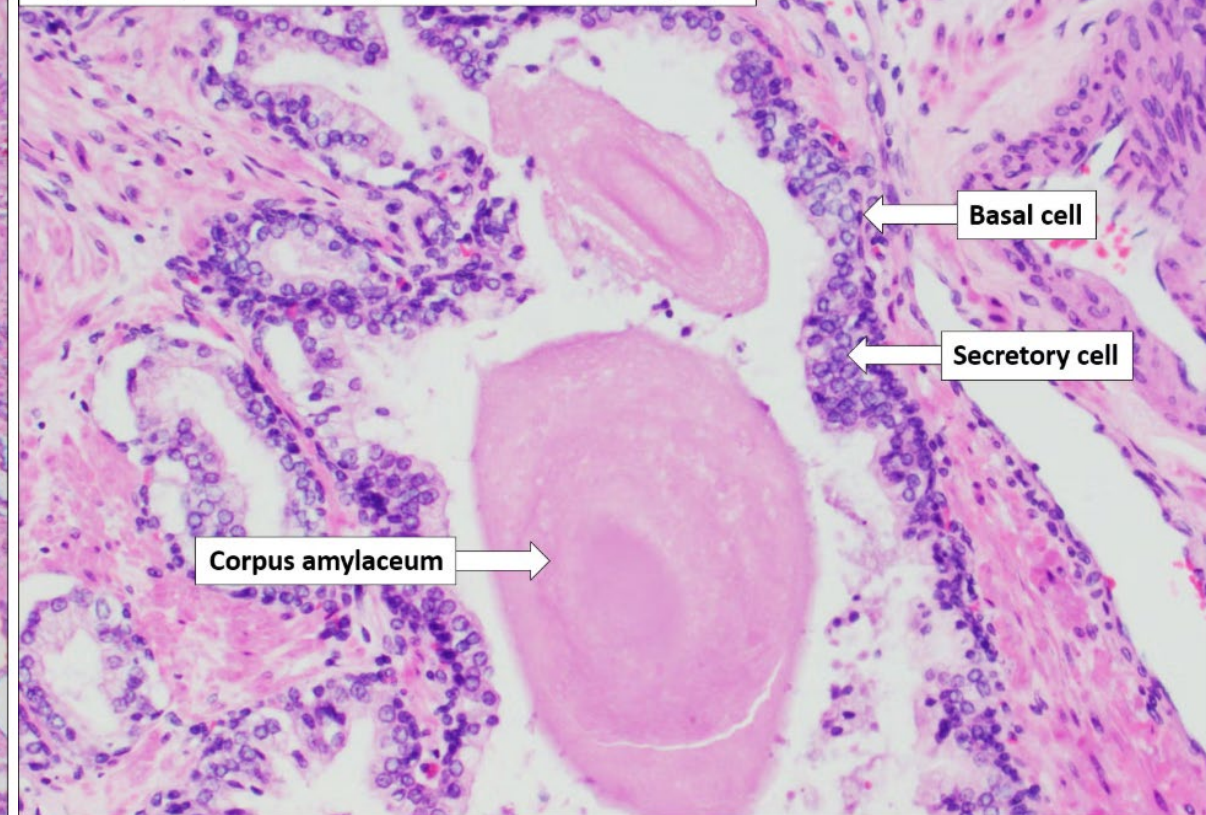
Prostate (low power):

The prostate comprises tubuloacinar glands within a fibromuscular stroma. Corpora amylacea may be seen within the glands.



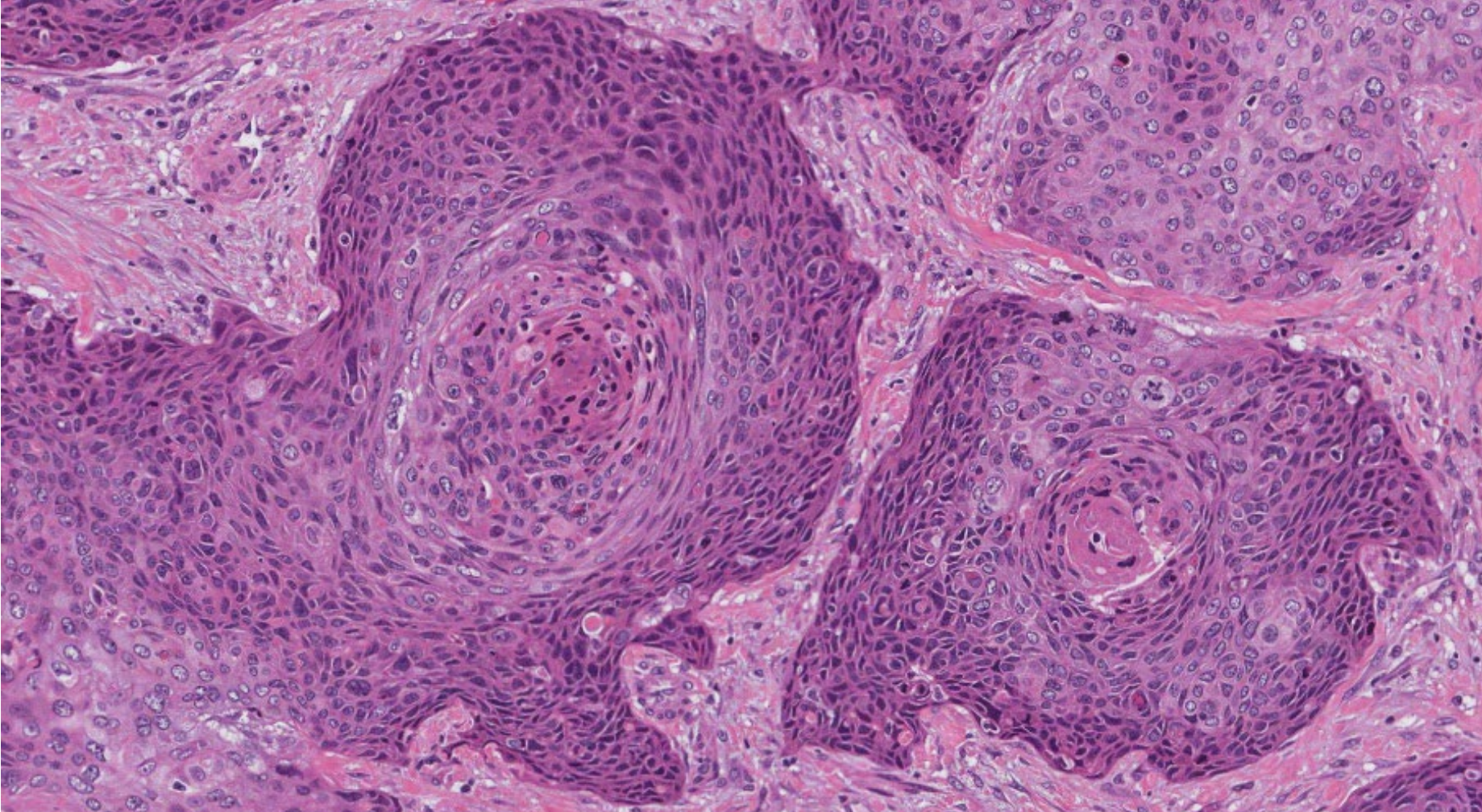
Prostate (high power):

Corpora amylacea are more prevalent with increasing age.



Station 6: Question 4, Penile SCC

Well-differentiated SCC with Keratin Pearls



**Station 7: Question 3, Match Histology
with Anatomical parts of the Male
reproductive organs**

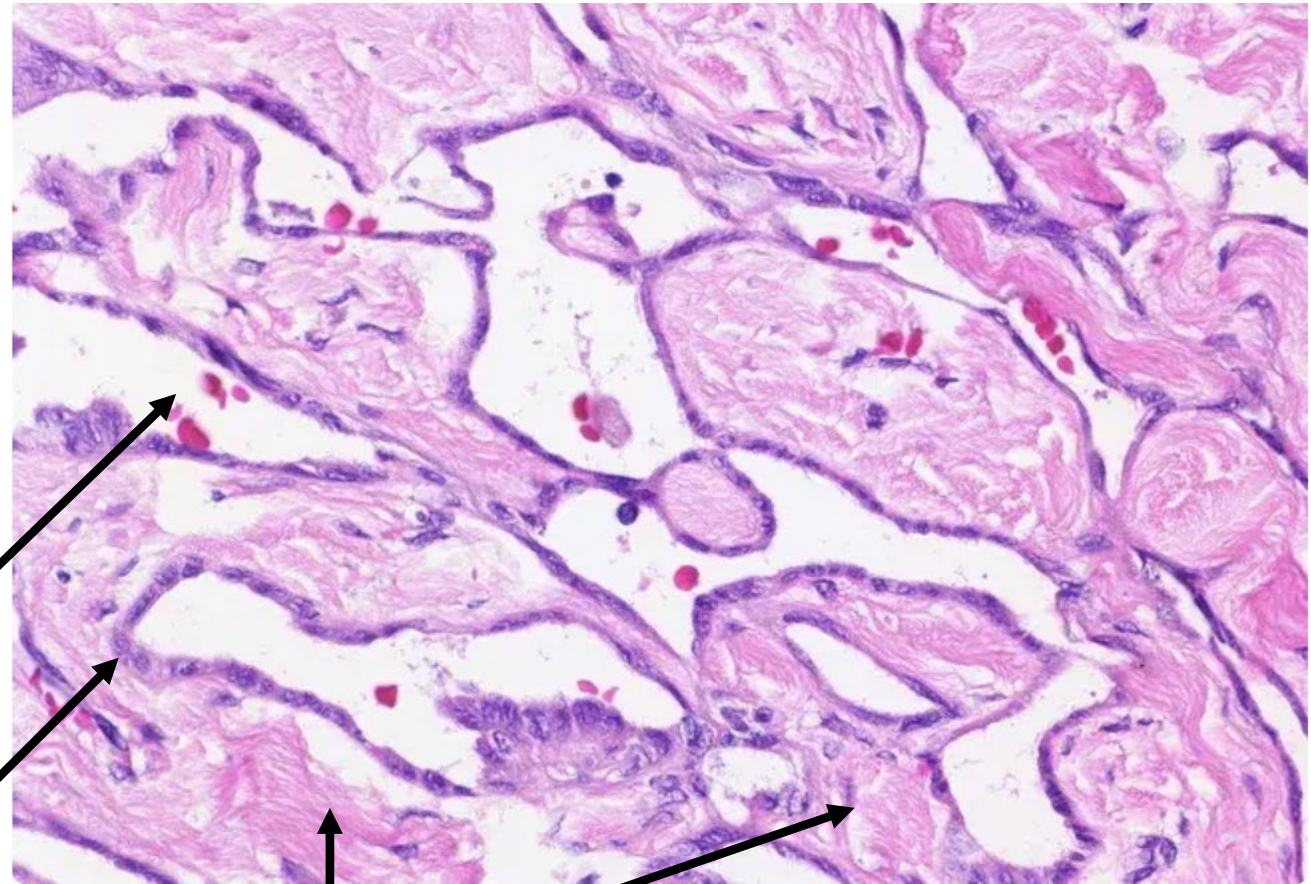
I. Rete Testis

- Rete testis is an **anastomosing network of delicate channels located in the testicular hilum**.
- It is lined by flattened squamous or low cuboidal epithelium. Epithelium-covered strands and columns seen in the photomicrograph are called chordae retis which connect the opposing walls of the channels.
- Rete receives the luminal contents of seminiferous tubules and empties them into the efferent ductules.

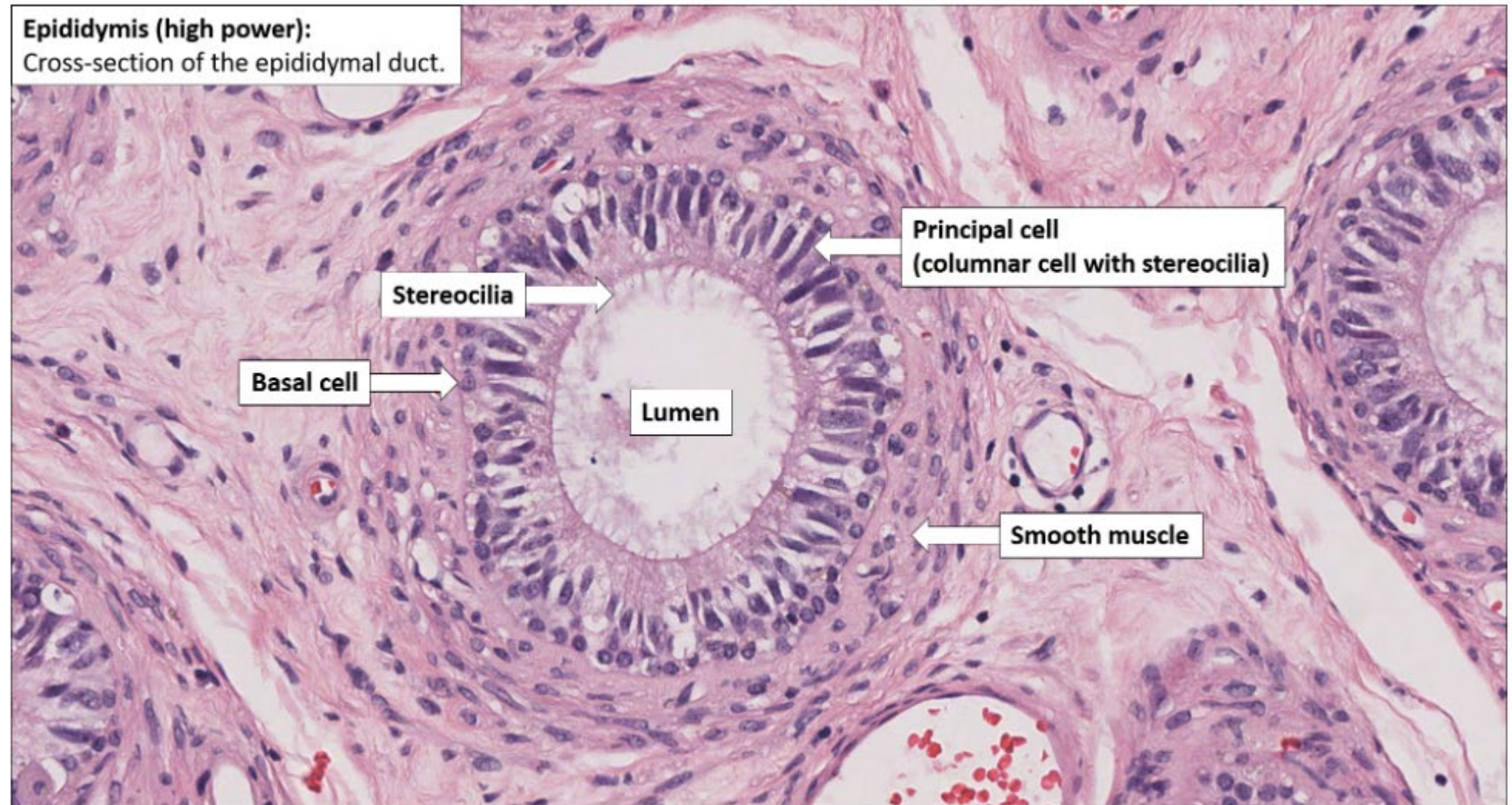
Network of channels

Epithelial cells covering the chordae retis

chordae retis



II. Epididymis



<https://medicine.nus.edu.sg/pathweb/normal-histology/testis-and-epididymis/>

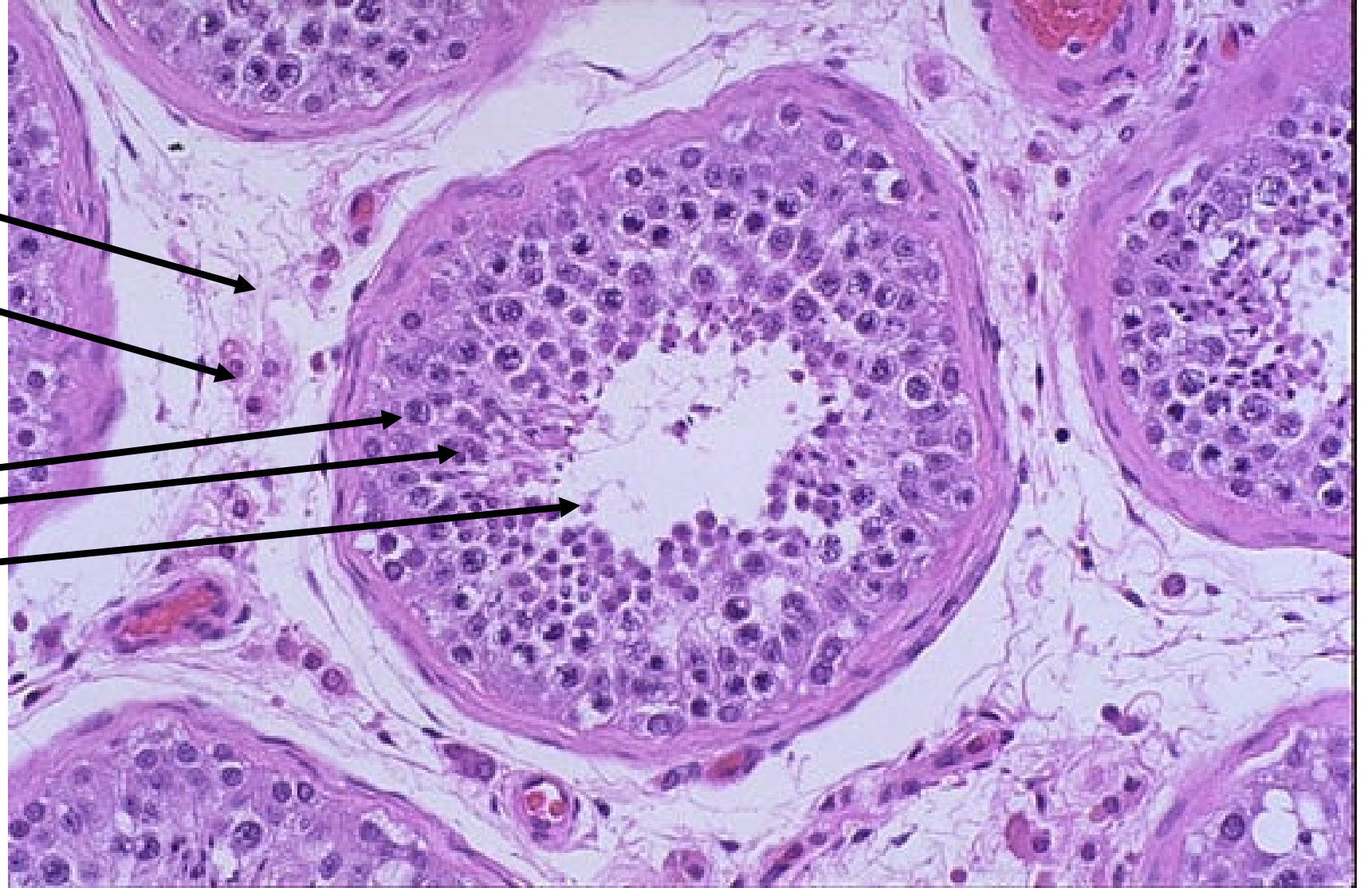
III. Ductus Deferens



IV. Seminiferous tubules For more images and annotations: https://medcell.org/histology/testis_sperm_development.php

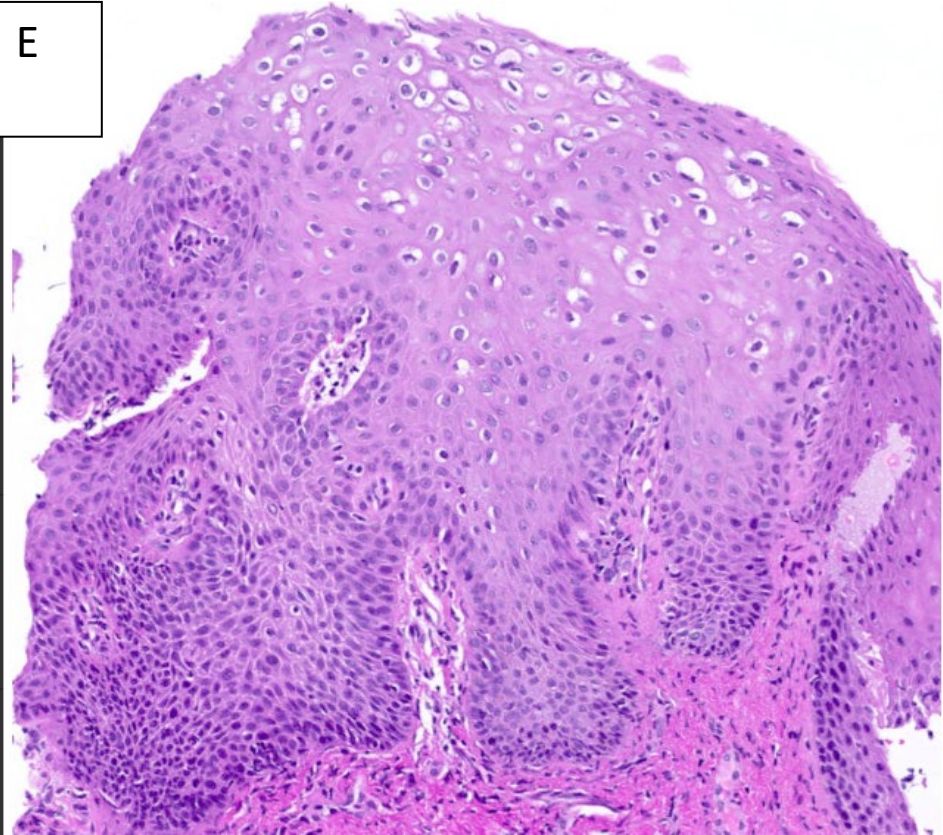
Normal adult testis is shown here at medium power, with a seminiferous tubule surrounded by **interstitium with Leydig cells**.

The tubule contains **primary** and **secondary** spermatogonia, and **spermatozoa**. The Sertoli (sustentacular) cells are indistinct because the germ cells are so numerous.



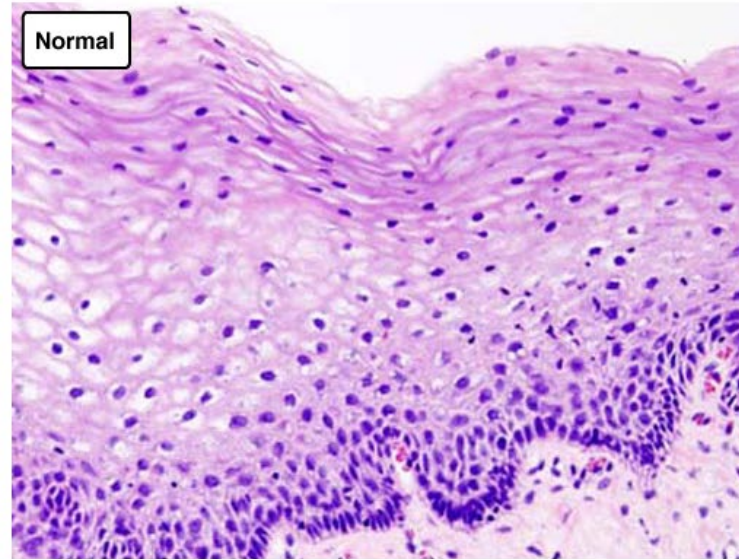
Cervical Histology

E

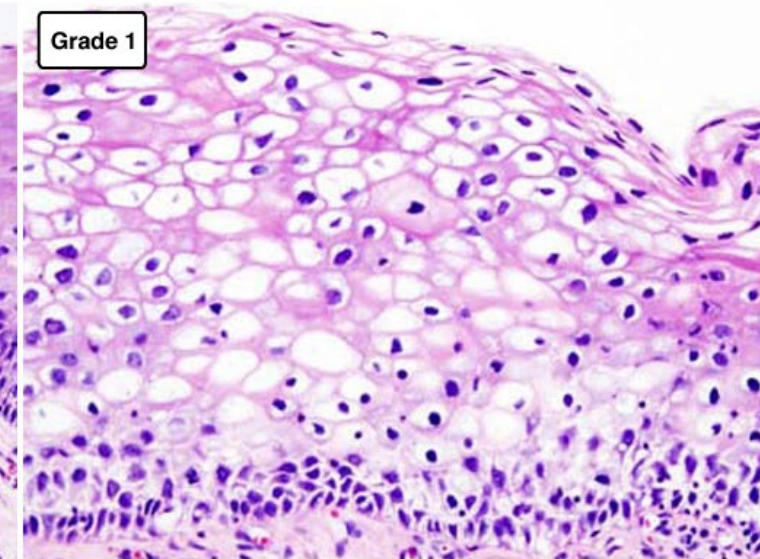


Condyloma with prominent koilocytosis (cells with perinuclear haloes) in upper cell layers.

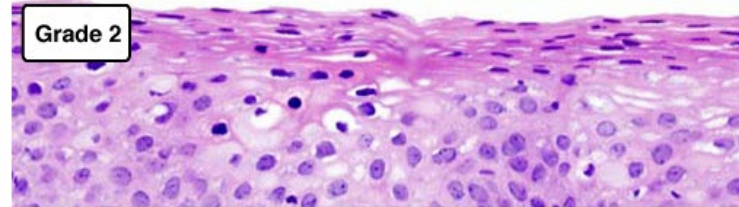
Normal



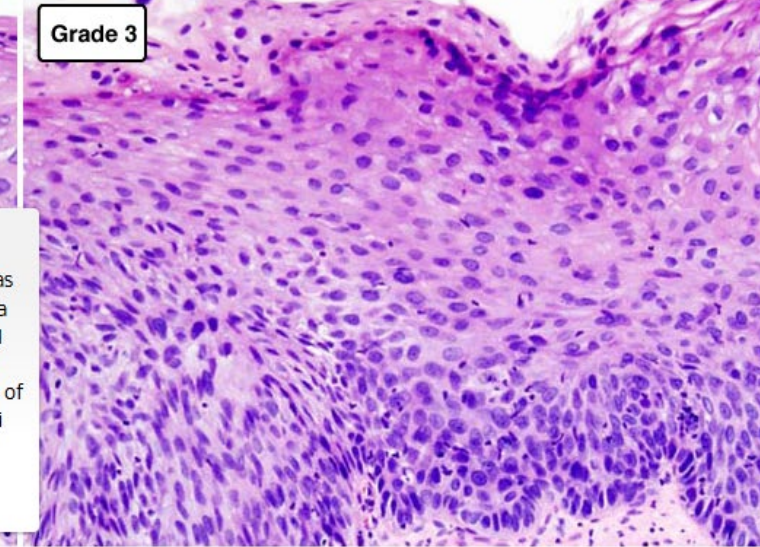
Grade 1



Grade 2



Grade 3

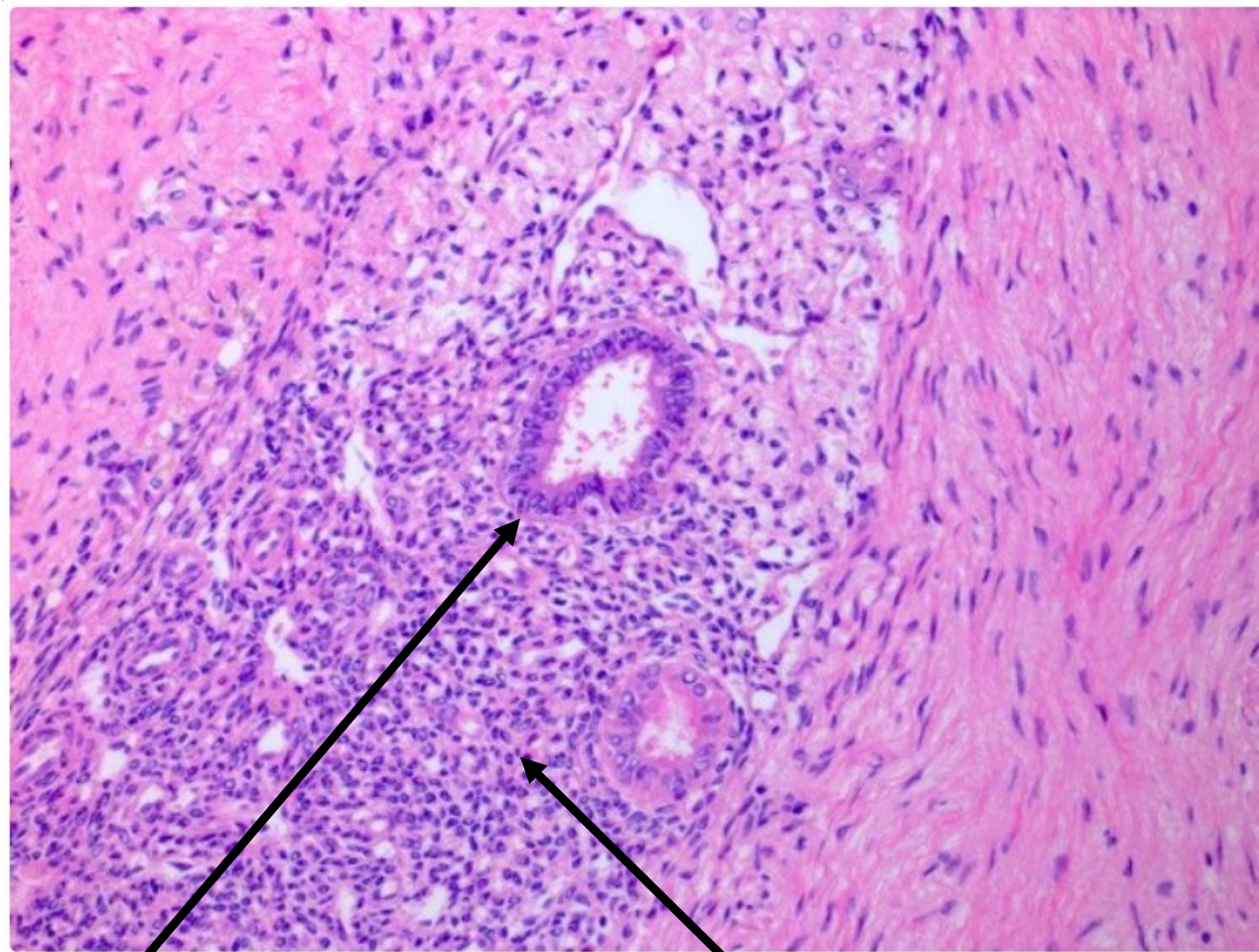


Cervical Intraepithelial Neoplasia

The cervix is located at the base of the uterus and is normally characterized as having a stratified squamous epithelium. In cervical intraepithelial neoplasia (CIN), the cells of the cervix undergo changes that reflect a loss of controlled growth. This begins in CIN I with mild dysplasia and nuclear angulation and vacuolization. The dysplasia in CIN II is more severe and affects more layers of the epithelium. It is important to note variation in the size of cells and nuclei during this phase, though the superficial layers of the epithelium may still appear normal. CIN III involves changes in cell and nuclear size, as well as abnormal mitoses, in all layers of the epithelium.

https://medcell.org/histology/epithelia_lab/cervical_intraepithelial_neoplasia.php

Station 8: Question 4, Endometriosis in Posterior Cul-De-Sac



Endometriosis with **endometrial type glands** and **endometrial stroma** .